MPG No. 32.3.120 $m'\bar{3}'m$ [Type III, cubic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss). * Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & M_{xxyy} & 0 & 0 & 0 \\ M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = 4\mathbb{T}^{(1)}_{xyz}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{yzyz} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{T}^{(2)}_{xyz}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{T}^{(3)}_{xyz}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & 0 & M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyy} & -M_{xxyy} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & -M_{yzyz} \end{bmatrix}$$

$$M_{xxyy} = 4\mathbb{T}^{(1)}_{xyz}$$
$$M_{yzyz} = \mathbb{T}^{(2)}_{xyz}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzyz} \end{bmatrix}$$

$$M_{yzyz} = \mathbb{T}^{(3)}_{xyz}$$

* Rank 4 tensor (t).